

אוניברסיטת בן-גוריון בנגב
Ben-Gurion University of the Negev

ICS OBSERVABLE DETECTION

DATA ENGINEERING PROJECT

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BACKGROUND

ICS

Industrial Control Systems

CTI Report

Cyber Threat Intelligence report

Observable

Any piece of information that can be used to detect or analyze an attack

MOTIVATION & UNIQUENESS

Detecting ICS attacks deep in OT layers

is still slow, manual, and error-prone.

Exploit OT-attack data

already exists in CTI reports that **no one has leveraged until now**

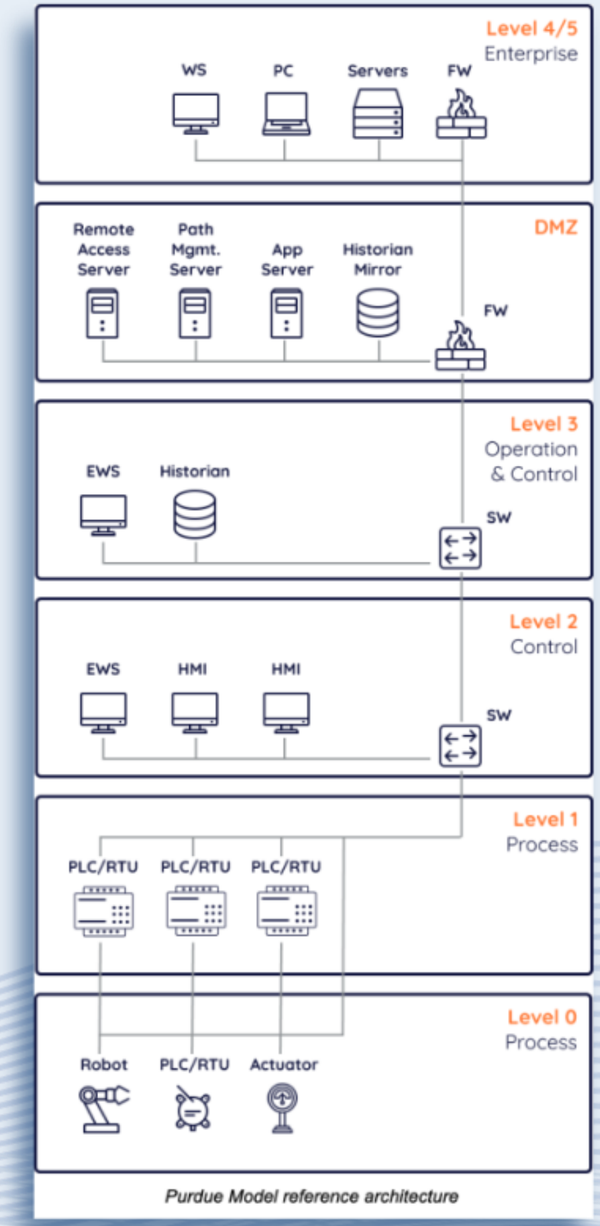
Automatic OT - observable extraction

establishes a detection baseline to flag **potential attacks**

PROBLEM

Within OT layers, observables are not as clearly defined as in IT layers





Purdue Model reference architecture



RESEARCH GOAL

Enhance protection across OT layers and expand information sharing within these layers



MAIN GOAL

ICS Observable Detection:

Identifying searchable artifacts withing CTI reports

CHALLENGES

- Define OT-layer observables
- Parsing PDFs
- Design goal-aligned LLM prompts
- Validate LLM results



DATA

73 MITRE ATT&CK® ICS PDFs
reports of real-world attacks

MITRE | ATT&CK®



SOLUTION FRAMEWORK

1. Data Collection & Pre-Processing
2. LLM Observable Extractor
3. LLM Rank Validator
4. Report Statistics

DATA COLLECTION

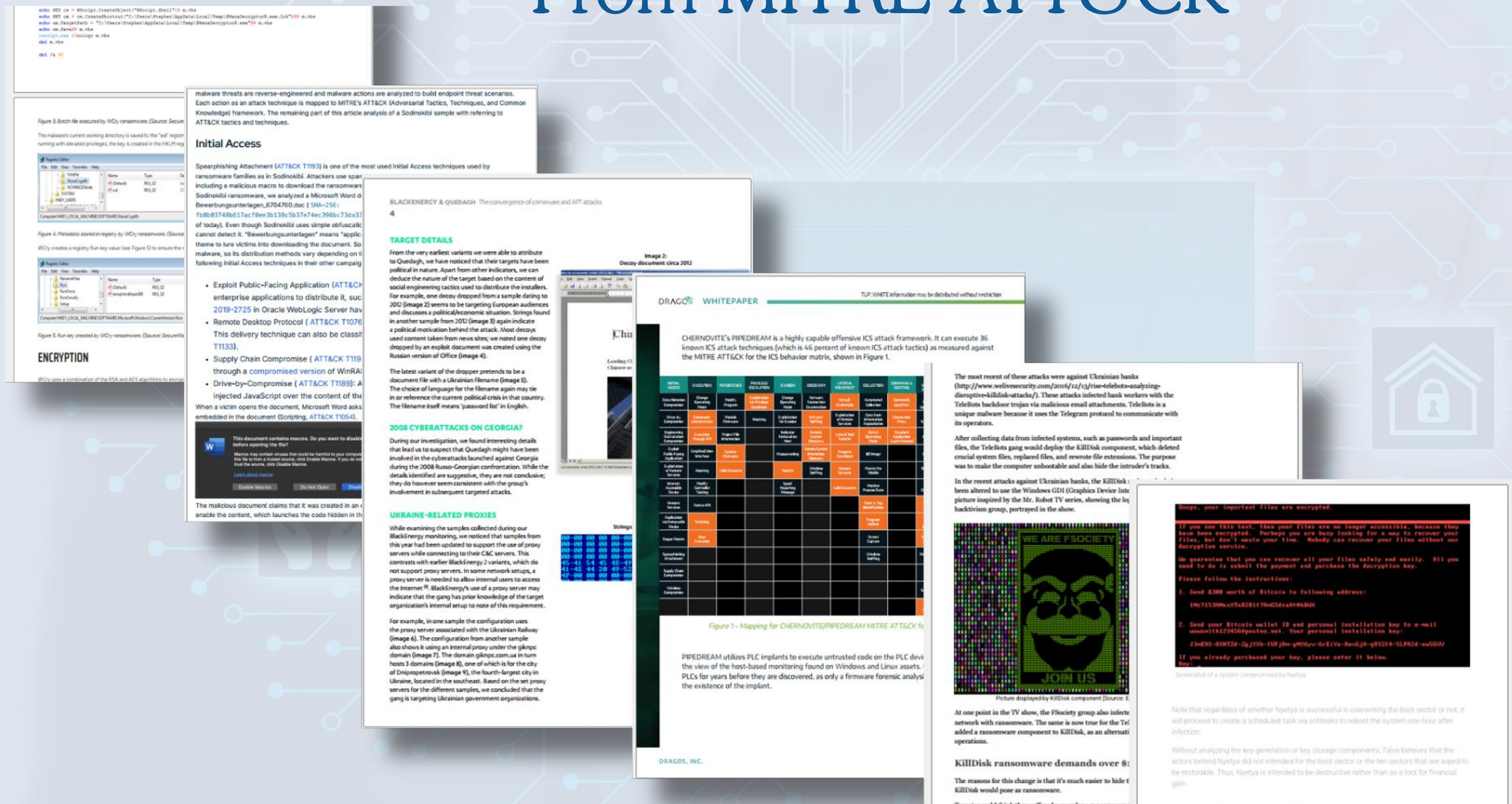
MITRE ATT&CK™

Extract observables from attack related procedure descriptions

	A	B	C	D	E	F	G	
1	observable_value	artifact_details	data_source	classification	STIX_supported	proprietary_artifact	parser	notes
2	Backdoor.Oldrea	Described	Malware analysis / Threat Intelligence	Malware	Full: Malware	Proprietary-Documented Technology		Payload referenced
3	Backdoor.Oldrea	Described	system logs	Malware	Full: Malware	Open/Standard Technology		Refers to a backdoc
4	Backdoor.Oldrea	Described	system logs	Software/Tool	Full: Malware	Proprietary-Documented Technology		Referenced with a M
5	Backdoor.Oldrea	Described	Threat Intelligence Feeds	Software/Tool	Full: Malware	Open/Standard Technology		Referenced via MIT
6	Backdoor.Oldrea ICS malware pl	Described	system logs	Software/Tool	Full: Malware	Proprietary-Documented Technology		Referenced with MI
7	Backdoor.Oldrea payload	Described	ICS data	Software/Tool	Full: Malware	Open/Standard Technology		Referenced via MIT
8	Backdoor.Oldrea RAT	Described	endpoint logs / threat intelligence feeds	Software/Tool	No	Proprietary-Undocumented Technology		Specific RAT with a
9	Backdoor.Oldrea RAT	Described	threat intelligence feeds, malware analysis reports	Software/Tool	Full: Malware	Open/Standard Technology		Identified via MITRE
10	C2	Mentioned	network logs	Network Entity	No	Open/Standard Technology		Generic reference to
11	CLSID	Mentioned	ICS data	Server Informati	No	Open/Standard Technology		Identifier for COM o
12	emails	Mentioned	email server logs	Email Message	Full: Email Message	Open/Standard Technology		Emails used as the
13	group count	Mentioned	ICS data	Server Informati	No	Open/Standard Technology		Number of groups c
14	OPC	Mentioned	ICS data, network logs	Industrial Comm	No	Open/Standard Technology		OPC is referenced g
15	OPC communications	Mentioned	ICS data	Communication	No	Open/Standard Technology		Generic reference to
16	OPC platforms	Mentioned	ICS data	Industrial Contro	No	Open/Standard Technology		General reference to
17	OPC server information	Mentioned	ICS Data	OPC Server	No	Open/Standard Technology		Generic server data
18	OPC servers	Mentioned	network logs	OPC Server	No	Open/Standard Technology		OPC (OLE for Proce
19	OPC tags	Mentioned	ICS Data	ICS Tag	No	Open/Standard Technology		The payload enume
20	OPC version	Mentioned	ICS data	Server Informati	No	Open/Standard Technology		Version information

SCRAPING 73 PDF REPORTS

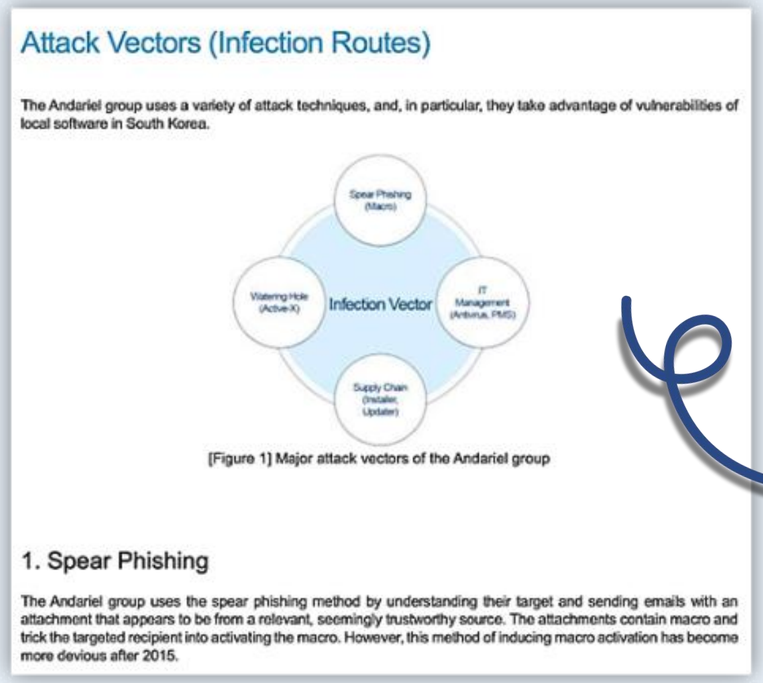
From MITRE ATT&CK



PREPROCESSING

CONVERT INTO .MD

Remove hyperlinks, annotations, headers, footers, hidden text & images (.Jpeg/.Jpg)



Attack Vectors (Infection Routes)

The Andariel group uses a variety of attack techniques, and, in particular, they take advantage of vulnerabilities of local software in South Korea.

[Figure 1] Major attack vectors of the Andariel group

1. Spear Phishing

The Andariel group uses the spear phishing method by understanding their target and sending emails with an attachment that appears to be from a relevant, seemingly trustworthy source. The attachments contain macro and trick the targeted recipient into activating the macro. However, this method of inducing macro activation has become more devious after 2015.

```
# Attack Vectors (Infection Routes)

The Andariel group uses a variety of attack techniques, and, in particular, they take advantage of vulnerabilities of local software in South Korea.

![](_page_3_Figure_3.jpeg)

[Figure 1] Major attack vectors of the Andariel group

## 1. Spear Phishing

The Andariel group uses the spear phishing method by understanding their target and sending emails with an attachment that appears to be from a relevant, seemingly trustworthy source. The attachments contain macro and trick the targeted recipient into activating the macro. However, this method of inducing macro activation has become more devious after 2015.

[Figure 2] below shows the contents of attachments used in the 2015 attacks with the original method of inducing macro activation. The user can enable macro, but there is not a significant need to do so because the content is visible.

![](_page_4_Picture_1.jpeg)

[Figure 2] Contents of documents used in the 2015 attack
```

OBSERVABLE EXTRACTION (.MD FILES)

- Split each .MD file into sections
  - **LLM-based** extraction prompt
    - **Identify Observables** - scan each section
      - **Assign Rank** - apply decision rules → Actionable, Described, Mentioned

EXTRACTION FORMAT

Extract every observable from CTI report snippets.

Ranks:

- Actionable – Detectable & unique (e.g., IPs, full URLs)
- Described – Incomplete or needs enrichment
- Mentioned – Conceptual, not usable for detection

Key Fields:

- Value, Rank, Source, Type
- STIX support, Tech type, Parser, Notes
- **fine & gross Classifications**

EXTRACTION RESULTS


```
{
  "observable_value": "McAfee ATR",
  "data_source": 'None",
  "STIX_supported": "No",
  "proprietary_artifact": "Proprietary-Documented Technology",
  "parser": "N/A",
  "notes": "Product/tool mentioned in the CTI report title.",
  "observable_id": 1,
  "artifact_details": "Mentioned"
  "fine_classification": "Software/Tool",
  "gross_classification": "Software"
},
```

VALIDATION RESULTS

OpenAI Agent - flags mis-ranked observables (CORRECT / INCORRECT)

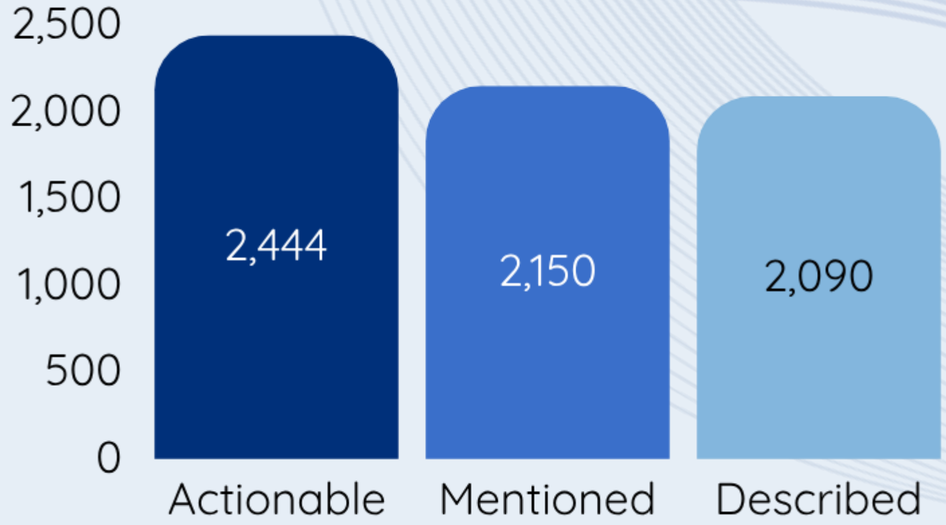
14	img → Described	CORRECT	As a template extracted from configuration, “img” is ambiguous and not unique enough to be considered actionable.
15	Subscribe → Described	CORRECT	“Subscribe” is a common keyword taken from configuration and lacks the uniqueness needed for an actionable indicator.
16	QUISERCENTRICS → Described	INCORRECT – should be "Actionable" because it is a complete and unique indicator string that can be directly matched by a detection rule.	It appears as a standalone, specific string from the configuration that affords low false-positive matching.
17	Cookiebot → Described	CORRECT	Although “Cookiebot” is a known product name, its use in this context is

95% Accuracy!

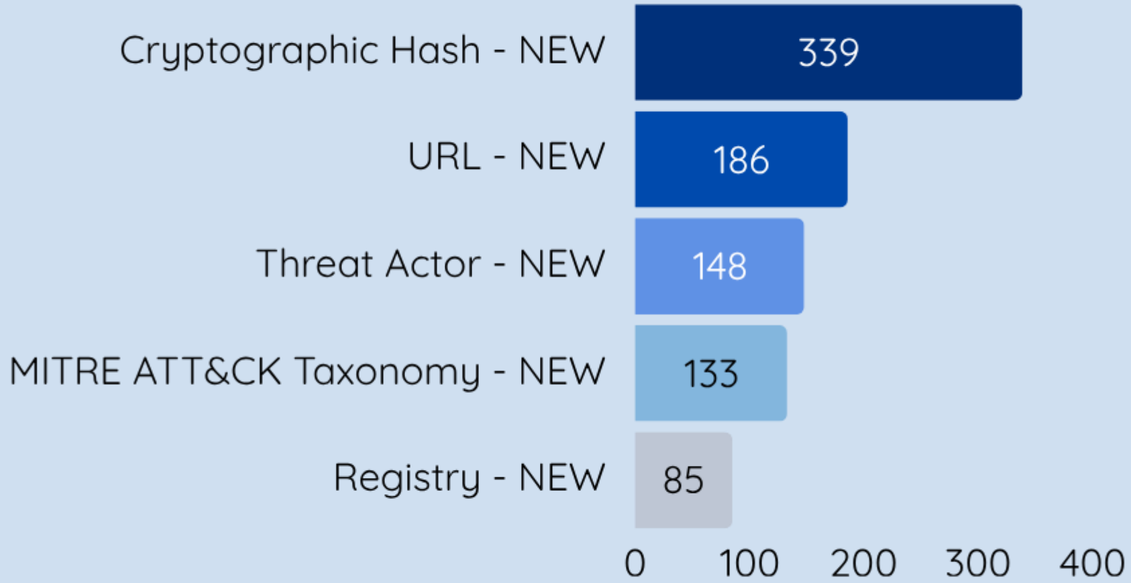

STATISTICS

Observable Distribution

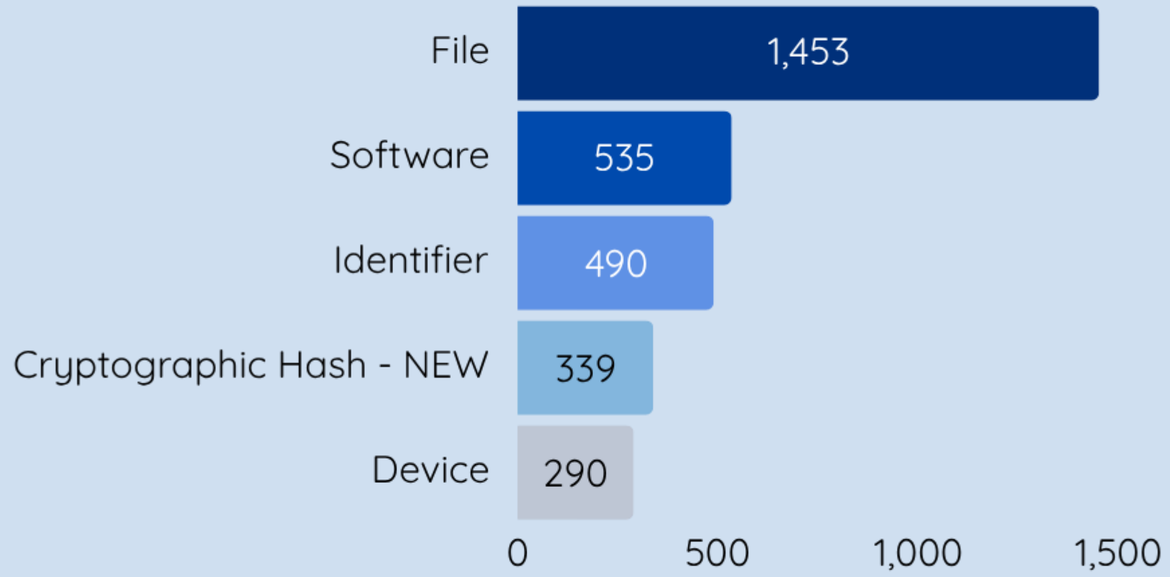
(By Category)



Top 5 “NEW” Categories

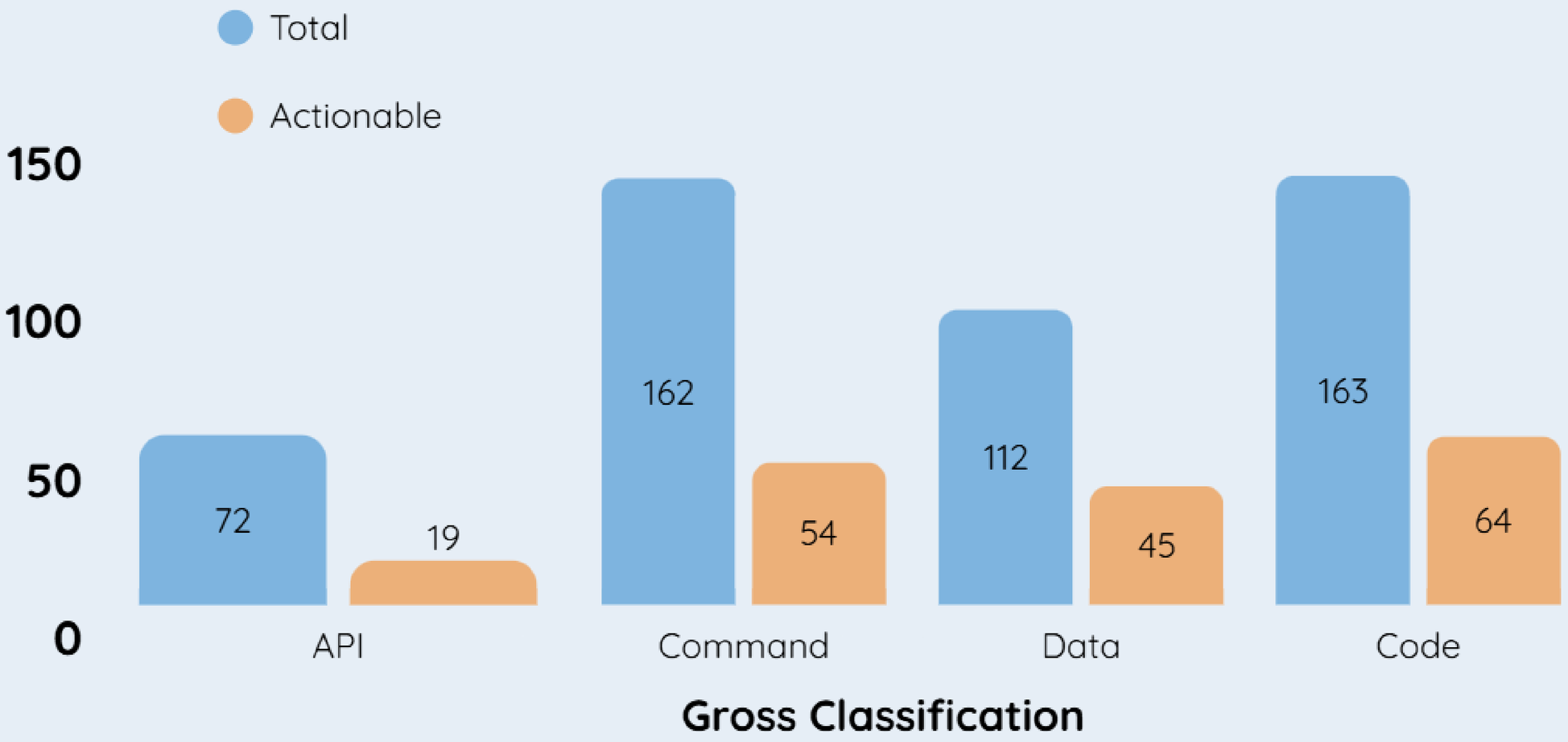


Top 5 Gross Classifications



Total VS Actionable

(By Category)



SUMMARY & CONCLUSIONS

Broad Coverage

Parsed ≈ 6,000 observables, with ~⅓ classified as Actionable - **most valuable for immediate defense**

Expanded Taxonomy - Added **179** new gross-classification labels

High Quality observable Extractor

End-to-end pipeline

transforms ICS-CTI reports into a metadata-rich dataset, enabling statistical analysis and data-driven insights

Thank You For Listening



Ben-Gurion University
of the Negev

GIT - Final Project - ICS Observable Detection